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Executive Summary

AI/ML Engineer specialising in **Generative AI, LLM systems, and MLOps**, with proven experience delivering production-grade machine learning solutions across NLP and computer vision. Expertise in **RAG pipelines, multi-modal sensor fusion, and scalable ML systems**, with a strong track record of improving model performance, optimising inference, and delivering measurable business impact. MSc Artificial Intelligence graduate with distinction, targeting AI Engineering, Applied AI, and research roles in the UK.

Technical Skills

Programming: Python, JavaScript, TypeScript

AI/ML: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers (BERT, RoBERTa, Longformer), GANs, LSTM, Reinforcement Learning

Generative AI: RAG, Prompt Engineering, LangChain, FAISS, Chroma, Hugging Face, LoRA, QLoRA

MLOps & Systems: Docker, CI/CD, MLflow, REST APIs, Git/GitHub, Linux, Redis

Cloud: AWS, GCP (fundamentals)

Domains: NLP, Computer Vision, SLAM/Odometry, Sensor Fusion, Medical AI, Quantum ML

Education

MSc Artificial Intelligence (Distinction: 7.8)

Sheffield, UK

Sheffield Hallam University

Jan 2025 – Mar 2025

Relevant Modules: Deep Learning, NLP, Computer Vision, Reinforcement Learning, AI Ethics

Capstone: Multi-modal sensor fusion for autonomous navigation with quantum-inspired optimisation

ET-I2oT Summer School (Distinction: 29/30)

Pisa, Italy

University of Pisa

Jul 2025 – Sep 2025

Focus: IoT systems, edge computing, distributed sensor networks

Selected Projects

Project Astraios: Quantum-Inspired Multi-Modal Visual Odometry (SHU–SJEC Collaboration)

Research Project — PyTorch — Computer Vision — Sensor Fusion

- Conducting research on multi-modal visual odometry integrating camera, LiDAR, and IMU using transformer architectures and temporal LSTM.
- Designed a quantum-inspired feature fusion approach, reducing trajectory drift by **15%** in low-texture environments.
- Achieved **0.99% trajectory error (36.4m RMSE over 3.6km)** across benchmark sequences.
- Built a scalable data pipeline with timestamp synchronisation and automated evaluation metrics.
- Results form the basis of an ongoing **research manuscript in preparation** in autonomous systems and robotics.

MedLinguists: Healthcare NLP + RAG for Patient QoL Analysis (NHS–SHU Collaboration)

NLP — Retrieval-Augmented Generation — Healthcare AI

- Developed an AI-driven patient experience analysis engine in collaboration with the NHS and Sheffield Hallam University.
- Processed **500+ patient narratives** to extract quality-of-life insights for abdominal wall hernia (AWH) patients.
- Designed NLP pipeline capturing psychological factors (body image, identity, mental health) often missed by conventional QoL tools.
- Built a RAG pipeline (FAISS + transformer models) improving clinical insight accuracy by **40%**.

- Generated structured reports for clinicians, reducing manual review time by **60%**.

Algonauts Challenge 2023: Brain-to-Image Reconstruction

GANs — Cognitive AI

- Built a conditional DCGAN model for reconstructing images from fMRI brain signals.
- Applied PCA (1024 → 128) and ResNet-50 feature conditioning for semantic alignment.
- Achieved **Top 10% ranking** in the challenge.
- Conducted evaluation using SSIM and LPIPS metrics on reconstructed outputs.

TravelAI: Review Clustering & Recommendation Engine

Unsupervised Learning — PCA — K-Means

- Clustered **5,456 user reviews** using a custom K-Means pipeline.
- Achieved an **87% silhouette score** (comparable to sklearn baseline).
- Reduced dimensionality by **60%** while preserving **95%** variance.
- Built a persona-based recommendation prototype, improving user engagement by **25%**.

Professional Experience

ML Engineer / Full-Stack Developer

Weboapp Discovery Pvt. Ltd.

Panchkula, India

Until Sep 2024

- Developed ML-powered recommendation and sentiment systems, increasing user retention by **30%**.
- Built and deployed ML pipelines using Docker and CI/CD, reducing release cycles by **50%**.
- Designed scalable REST APIs with Redis caching, improving response times by **35%**.
- Led frontend migration to React + TypeScript, reducing production bugs by **40%**.

Research Interests

Multi-Modal Sensor Fusion, Autonomous Systems, Computer Vision, Generative AI, Quantum-Inspired Machine Learning, Healthcare AI, Large Language Models (LLMs)

Awards & Achievements

- **Winner – Robot Hackathon (RoboSapiens Team)**
Awarded at Sheffield Hallam University Festival of Computing for developing an innovative robotics solution.
- **Excellence Certificate – Project Astraios (SHU–SJEC Collaboration)**
Recognised for outstanding technical contribution in an international AI/engineering project.

Publications (In Preparation)

- **Singh, G.** *Quantum-Inspired Multi-Modal Visual Odometry*. Manuscript in preparation (2026).

Additional Information

Languages: English, Hindi, Punjabi

Interests: Quantum AI, Edge AI, Responsible AI

Community: Environmental initiative – planted 500+ trees

References

Available upon request